

Introduction to General Relativity: Homework 8

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Problem 1. Consider the Robertson-Walker spacetime described by the metric

$$ds^2 = dt^2 - e^{2t/t_H} [dr^2 + r^2 (d\theta^2 + \sin^2(\theta) d\varphi^2)],$$

with the coordinates $x^\mu = (t, r, \theta, \varphi)$, where t is the universal time and $t_H \approx 14$ Gyr is the Hubble time.

- (i) (5 points) Compute the path $r(t)$ of a light pulse emitted at time $t = t_0$ at the origin $r = 0$. You may assume that the light ray moves along the line $\theta = \pi/2$ and $\varphi = 0$ so that $dr/dt > 0$.
- (ii) (5 points) Compute the proper distance between the origin $r = 0$ and a point $r > 0$ on the line $\theta = \pi/2$ and $\varphi = 0$ at a fixed universal time t .
- (iii) (5 points) Compute the spectral shift of the light pulse in (i) during the time between emission at the origin and arrival at a point $r > 0$.

Hint: the spectral shift is defined as $z = \lambda_{rec}/\lambda_{em} - 1$, where λ_{em} is the wavelength of the light at emission and λ_{rec} its wavelength when it arrives.

Solution to (i). For a light ray we have a null curve, so $ds^2 = 0$. Since the photon moves along

$$\theta = \frac{\pi}{2}, \quad \varphi = 0,$$

we have $d\theta = 0$ and $d\varphi = 0$. The metric therefore reduces to

$$0 = dt^2 - e^{2t/t_H} dr^2.$$

Thus

$$dt^2 = e^{2t/t_H} dr^2,$$

and taking the positive root (since $dr/dt > 0$),

$$\frac{dr}{dt} = e^{-t/t_H}.$$

Integrating from emission time t_0 at $r = 0$,

$$r(t) = \int_{t_0}^t e^{-t'/t_H} dt'.$$

Compute the integral:

$$\int e^{-t'/t_H} dt' = -t_H e^{-t'/t_H}.$$

Hence

$$r(t) = \left[-t_H e^{-t'/t_H} \right]_{t_0}^t = t_H \left(e^{-t_0/t_H} - e^{-t/t_H} \right)$$

Thus the path of the light pulse is

$$r(t) = t_H \left(e^{-t_0/t_H} - e^{-t/t_H} \right). \quad \square$$

Solution to (ii). The proper distance at fixed universal time t is obtained from the spatial part of the metric with $dt = 0$.

Along the line $\theta = \pi/2$ and $\varphi = 0$ we again have $d\theta = d\varphi = 0$, so

$$ds^2 = -e^{2t/t_H} dr^2.$$

The proper spatial distance is therefore

$$d\ell = e^{t/t_H} dr.$$

The proper distance between $r = 0$ and r at fixed t is

$$\ell(t, r) = \int_0^r e^{t/t_H} dr'.$$

Since t is fixed during the integration,

$$\ell(t, r) = e^{t/t_H} r.$$

Thus the proper distance is

$$\ell(t, r) = e^{t/t_H} r. \quad \square$$

Solution to (iii). In a Robertson–Walker spacetime the wavelength of light scales with the scale factor $a(t)$. From the metric we identify

$$a(t) = e^{t/t_H}.$$

Therefore

$$\frac{\lambda_{rec}}{\lambda_{em}} = \frac{a(t_{rec})}{a(t_{em})}.$$

If the light is emitted at $t_{em} = t_0$ and received at $t_{rec} = t$, then

$$\frac{\lambda_{rec}}{\lambda_{em}} = \frac{e^{t/t_H}}{e^{t_0/t_H}} = e^{(t-t_0)/t_H}.$$

The spectral shift is defined as

$$z = \frac{\lambda_{rec}}{\lambda_{em}} - 1,$$

so

$$z = e^{(t-t_0)/t_H} - 1.$$

Thus the redshift of the photon is

$$z = e^{(t-t_0)/t_H} - 1. \quad \square$$

Problem 2. A scalar field φ with potential energy density $V(\varphi)$ has a Lagrangian density given by

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} (\partial_\mu \varphi) (\partial_\nu \varphi) - V(\varphi).$$

- (i) (5 points) Derive the equation of motion for the scalar field φ .
- (ii) (10 points) Assuming that the $N + 1$ -dimensional spacetime has a metric given by

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = dt'^2 - a(t)^2 G_{ij} dx^i dx^j,$$

where G_{ij} are the metric components on an N -dimensional Riemannian manifold, and that the scalar field φ only depends on the time coordinate t , show that the scalar field is an ideal fluid and find the (time-independent) equation-of-state parameter $w = p/\rho_0$.

Solution to (i). The equation of motion follows from the Euler–Lagrange equation for a scalar field,

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} \right) = 0.$$

First compute

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \varphi)} = g^{\mu\nu} \partial_\nu \varphi.$$

Next compute

$$\frac{\partial \mathcal{L}}{\partial \varphi} = -V'(\varphi).$$

Substituting into the Euler–Lagrange equation gives

$$-V'(\varphi) - \partial_\mu (g^{\mu\nu} \partial_\nu \varphi) = 0.$$

Thus the equation of motion is

$$\partial_\mu (g^{\mu\nu} \partial_\nu \varphi) - V'(\varphi) = 0.$$

In curved spacetime it is conventional to write this using the covariant d'Alembertian,

$$\square \varphi = \nabla_\mu \nabla^\mu \varphi,$$

so the Klein–Gordon equation becomes

$$\nabla_\mu \nabla^\mu \varphi - V'(\varphi) = 0. \quad \square$$

Solution to (ii). The energy–momentum tensor for a scalar field is

$$T_{\mu\nu} = \partial_\mu \varphi \partial_\nu \varphi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi - V(\varphi) \right).$$

Assume the scalar field depends only on time:

$$\varphi = \varphi(t).$$

Thus

$$\partial_i \varphi = 0, \quad \partial_t \varphi = \dot{\varphi}.$$

First compute the kinetic term

$$g^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi = g^{tt} \dot{\varphi}^2 = \dot{\varphi}^2.$$

Therefore

$$\mathcal{L} = \frac{1}{2} \dot{\varphi}^2 - V(\varphi).$$

The energy density is $\rho = T_{tt}$. Using the expression for $T_{\mu\nu}$,

$$T_{tt} = \dot{\varphi}^2 - g_{tt} \left(\frac{1}{2} \dot{\varphi}^2 - V(\varphi) \right).$$

Since $g_{tt} = 1$, we obtain

$$\rho = \frac{1}{2} \dot{\varphi}^2 + V(\varphi).$$

For pressure, the spatial components of the energy–momentum tensor are

$$T_{ij} = -g_{ij} \left(\frac{1}{2} \dot{\varphi}^2 - V(\varphi) \right).$$

Since an ideal fluid has

$$T_{ij} = p g_{ij},$$

we identify

$$p = \frac{1}{2} \dot{\varphi}^2 - V(\varphi).$$

Thus the scalar field behaves exactly like an ideal fluid with

$$\rho = \frac{1}{2} \dot{\varphi}^2 + V(\varphi), \quad p = \frac{1}{2} \dot{\varphi}^2 - V(\varphi).$$

The equation-of-state parameter is

$$w = \frac{p}{\rho}.$$

Hence

$$w = \frac{\frac{1}{2} \dot{\varphi}^2 - V(\varphi)}{\frac{1}{2} \dot{\varphi}^2 + V(\varphi)}.$$

This parameter is constant in time if the ratio between the kinetic energy and potential energy remains constant. $\square$